UNIT III

Static Testing: Inspections, Structured Walkthroughs, Technical Reviews.

Validation activities: Unit testing, Integration Testing, Function testing, system testing, acceptance testing.

Regression testing: Progressives Vs regressive testing, Regression test ability, Objectives of regression testing, Regression testing types, Regression testing techniques.

Static Testing

Features of Software Testing:

- Static testing techniques do not demonstrate that the software is operational or one function of software is working.
- It checks the software product at each SDLC stage for conformance with the required specifications or standards. Requirements, design specifications, test plans, source code, user's manuals, maintenance procedures are some of the items that can be statically tested.
- Static testing has proved to be a cost-effective technique of error detection.
- Another advantage in static testing is that a bug is found at its exact location whereas a bug found in dynamic testing provides no indication to the exact source code location.

Types of Static Testing

1. Software Inspections
2. Walkthroughs
3. Technical Reviews

1. Inspections

- Inspection process is an in-process manual examination of an item to detect bugs.
- Inspection process is carried out by a group of peers. The group of peers first inspects the product at individual level. After this, they discuss potential defects of the product observed in a formal meeting.
- It is a very formal process to verify a software product. The documents which can be inspected are SRS, SDD, Code and test plan.
- Inspection process involves the interaction of the following elements:
    - a) Inspection steps
    - b) Roles for participants
    - c) Item being inspected

1.1 Inspection Team

For the inspection process, a minimum of the following four team members are required.

Author / Owner / Producer:

- A programmer or designer responsible for producing the program or document.
- He is also responsible for fixing defects discovered during the inspection process.

Inspector:

- A peer member of the team, i.e he is not a manager or supervisor.
- He is not directly related to the product under supervision and may be concerned with some other products.

Moderator:

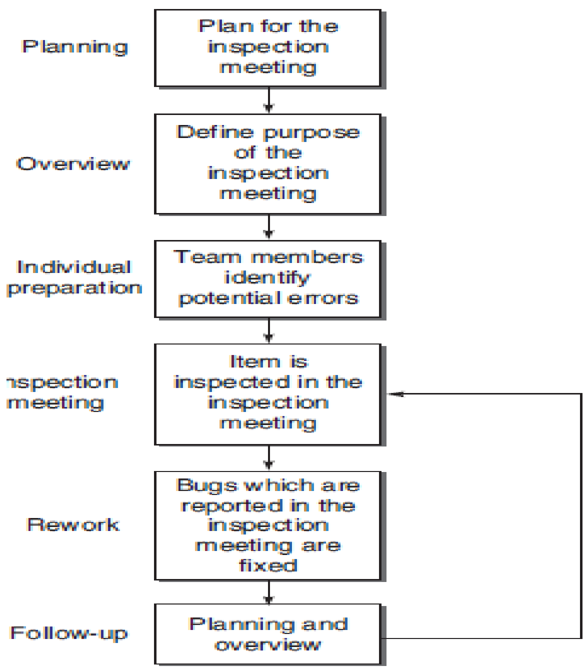
- A team member who manages the whole inspection process.
- He schedules, leads, and controls the inspection session.

Recorder:

- One who records all the results of the inspection meeting.

1.1.2 Inspection Process

A general inspection process has the following stages



Planning

During this phase the following is executed:

- The product to be inspected is being identified.
- A moderator is assigned.
- The objective of the inspection is stated i.e whether the inspection is to be conducted for defect detection or something else.

During planning, the moderator performs the following activities:

- Assures that the product is ready for inspection.
- Selects the inspection team and assigns their roles.
- Schedules the meeting venue and time.
- Distributes the inspection material like the item to be inspected, client lists etc.

Overview

- In this stage, the inspection team is provided with the background information for inspection.
- The author presents the rationale (reasons) of the product, its relationship to the rest of the products being developed, its function and intended use and the approach used to develop it.

Individual Preparation

- After the overview, the reviewers individually prepare themselves for the inspection process by studying the documents provided to them in the overview session.
- They point out potential errors or problems found and record in a log. This log is then submitted to the moderator.
- The moderator compiles the logs of different members and gives a copy of this compiled list to the author of the inspected item.

Inspection Meeting

- Once all the initial preparation is complete, the actual inspection meeting can start.
- The inspection meeting starts with the author of the inspected item who has created it.
- The author first discusses every issue raised by different members in the compiled log file.
- After the discussion, all the members arrive at a consensus (agreement) whether the issues pointed out are in fact errors and if they are errors, should they be admitted by the author.

Rework

- The summary list of the bugs that arise during the inspection meeting needs to be reworked by the author.
- The author fixes all these bugs and reports back to the moderator.

Follow-Up

- It is the responsibility of the moderator to check that all the bugs found in the last meeting have been resolved.
- The document is then approved for release.

1.1.3 Benefits of Inspection Process

Bug reduction

- The number of bugs is reduced through the inspection process reported by the inspection process in IBM.
- The number of bugs per thousand lines of code has been reduced by two-thirds.

Bug prevention

- Inspections can also be used for bug prevention.
- Based on the experience of previous inspections, analysis can be made for future inspections or projects, thereby preventing the bugs which have appeared earlier.

Productivity

- Since all phases of SDLC may be inspected without waiting for code development and its execution, the cost of finding bugs decreases and increases productivity.

Real-time feedback to software engineers

- Software developers get feedback on their products on a relatively real-time basis. Developers find out the type of mistakes they make and what is the error density.
- Since they get this feedback in the early stage of the development, they may improve their capability.

Reduction in development resource

- The cost of rework is high if inspections are not used and errors are found during development or testing.
- Therefore, techniques should be adopted such that errors are found and fixed to their place of origin as possible.
- Inspections reduce the effort required for dynamic testing and any rework during design and code, thereby causing an overall net reduction in the development resource.

Quality improvement

- The direct consequence of static testing also results in the improvement of quality of the final product.
- Inspections help to improve the quality by checking the standard compliance, modularity, clarity, and simplicity.

Project management

- A project needs monitoring and control. It depends on some data obtained from the development team.
- Inspection is another effective tool for monitoring the progress of the project.

Checking coupling and cohesion

- The modules' coupling (interdependencies between modules) and cohesion (how related the functions within a single module) can be checked easily through inspection as compared to dynamic testing.
- This also reduces the maintenance work.

Learning through inspection

- Inspection also improves the capability of different as they learn various types of bugs and the reasons why they occur through discussions.
- It may be more beneficial for new members. They can learn about the project in a very short time.
- This helps them in the later stages of development and testing.

Process improvement

- There is always scope of learning from the results of one process. They analyze why the errors occurred or the frequent places where the errors occurred can be done by the inspection team members.
- The analysis results can then be used to improve the inspection process so that the current as well as future projects can benefit.

Finding most error-prone modules

- Through the inspection process, the modules can be analyzed based on the error-density of the individual module.

Module Name	Error density (Error/ KLoC)
A	23
B	13
C	45

Error-prone modules

- From the example modules given in the above table Module C is more error prone. This information can be used and some decision should be taken as to:
    i. Redesign the module
    ii. Check and rework on the code of Module C
    iii. Take extra precautions and efficient test cases to test the module

Distribution of error-types

- Inspections can also be used to provide data according to the error-types.
- It means that we can analyze the data of the percentage of bugs in a particular type of bug category.

Bug Type	No. of errors	%
Design Bugs	78	57.8
Interface Bugs	45	33.4
Code Bugs	12	8.8

Bug type distribution

- If we get this data very early in the development process, then we can analyze which particular type of bugs is repeating. We can modify the process well in time and the results can also be used in future.

1.1.4 Effectiveness of Inspection Process

- The effectiveness of the inspection process lies in its rate of inspection.
- The rate of inspection refers to how much evaluation of an item has been done by the team.
- If the rate is very fast, it means the coverage of item to be evaluated is high. Similarly if the rate is too slow, it means that the coverage is not much.
- The rate should also be considered in the perspective of detection of errors.
- If the rate is too fast, then it may be possible that it reveals only few errors. If the rate is too slow, then the cost of project increases.
- Error detection efficiency of the inspection process, as given below:

$$\text{Error detection efficiency} = \frac{\text{Error found by an inspection}}{\text{Total errors in the team before inspection}} * 100$$

1.1.5 Cost of Inspection Process

- At least four members involved in an inspection team, the cost of inspecting 100 lines of code is roughly equivalent to one person-day of effort.
- This assumes that the inspection itself takes about an hour and that each member spends 1-2 hours preparing for the inspection.
- Testing costs depend on the number of faults in the software. However, the effort required for the program inspection is less than half the effort that would be required for equivalent dynamic testing.
- It has been estimated that the cost of inspection can be 5–10% of the total cost of the project.

1.1.6 Variants of Inspection Process

The following lists some of the variants of the formal inspection.

Active Design Reviews (ADRs)	Several reviews are conducted targeting a particular type of bugs and conducted by the reviewers who are experts in that area.
Formal Technical Asynchronous review method (FTArm)	Inspection process is carried out without really having a meeting of the members. This is a type of asynchronous inspection in which the inspectors never have to simultaneously meet.
Gilb Inspection	Defect detection is carried out by individual inspector at his level rather than in a group.
Humphrey's Inspection Process	Preparation phase emphasizes the finding and logging of bugs, unlike Fagan inspections. It also includes an analysis phase wherein individual logs are analysed and combined into a single list.
N-Fold inspections	Inspection process's effectiveness can be increased by replicating it by having multiple inspection teams.
Phased Inspection	Phased inspections are designed to verify the product in a particular domain by experts in that domain only.
Structured Walkthrough	Described by Yourdon. Less formal and rigorous than formal inspections. Roles are coordinator, scribe, presenter, reviewers, maintenance oracle, standards bearer, user representative. Process steps are Organization, Preparation, Walkthrough, and Rework. Lacks data collection requirements of formal inspections.

Active Design Reviews

- Active Design Reviews are for inspecting the design stage of SDLC.
- In this process, several reviews are conducted targeting a particular type of bugs and conducted by the reviewers who are experts in that area.
- It covers all the sections of the design document based on several small reviews instead of only one inspection.
- Reviews based on the use of questionnaires to give the reviewers better-defined responsibilities and to make them play a more active role.

Active design reviews process

Overview

- A brief overview of the module being reviewed is presented.
- The overview explains the modular structure in case it is unfamiliar to reviewers, and shows them where the module belongs in the structure.

Review

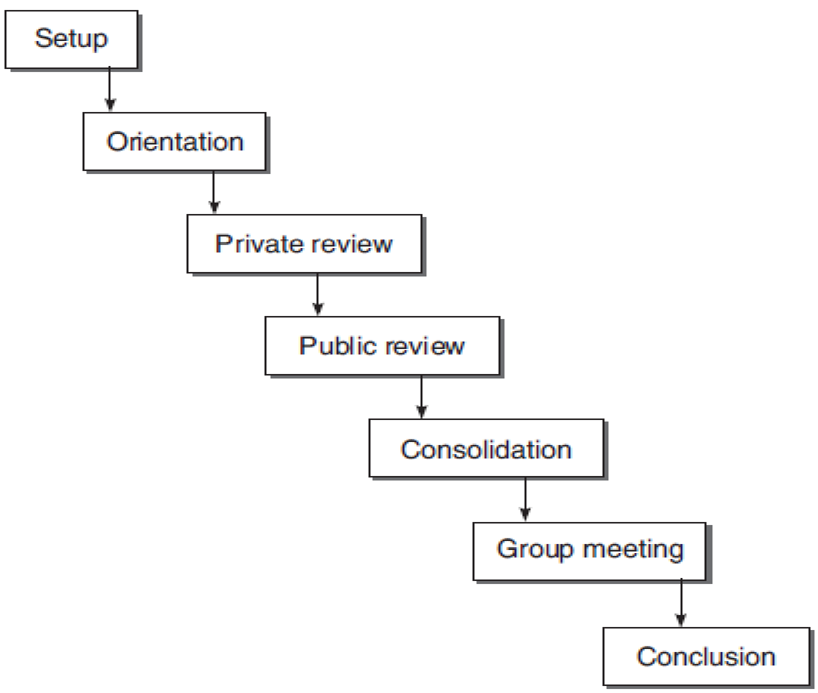
- Reviewers are assigned sections of the document to be reviewed and questionnaires based on the bug type.

Meeting

- The designers read the completed questionnaires and meet the reviewers to resolve any queries that the designers may have about the reviewer's answers to the questionnaires.
- The reviewer's answers to the questionnaires. defend their answers. This interaction continues until both designers and reviewers understand the issues, an agreement on these issues may not be reached.
- After the review, the designers produce a new version of the documentation.

Formal Technical Asynchronous Review Method (FTArm)

- In this process, the meeting phase of inspection is considered expensive and therefore, the idea is to eliminate this phase.
- The inspection process is carried out without having a meeting of the members. This is a type of asynchronous inspection in which the inspectors never have to simultaneously meet.
- For this process, an online version of the document is made available to every member where they can add their comments and point out the bugs.
- This process consists of the following steps as shown below:



Asynchronous method

Setup

- It chooses the members and prepare the hypertext document for an asynchronous inspection process.

Orientation

- It is same as the overview step of the inspection process.

Private review

- It is same as the preparation phase in the inspection process.
- Here, each reviewer or inspector individually gives his comments on the document being inspected individually and is unable to see the other inspector's comments.

Public review

- In this step, all comments provided privately are made public and all inspectors are able to see each other's comments and put forward their suggestions.
- Based on this feedback and with consultation of the author, it is decided that there is a bug.

Consolidation

- In this step, the moderator analyses the result of private and public reviews and lists the findings and unresolved issues, if any.

Group meeting

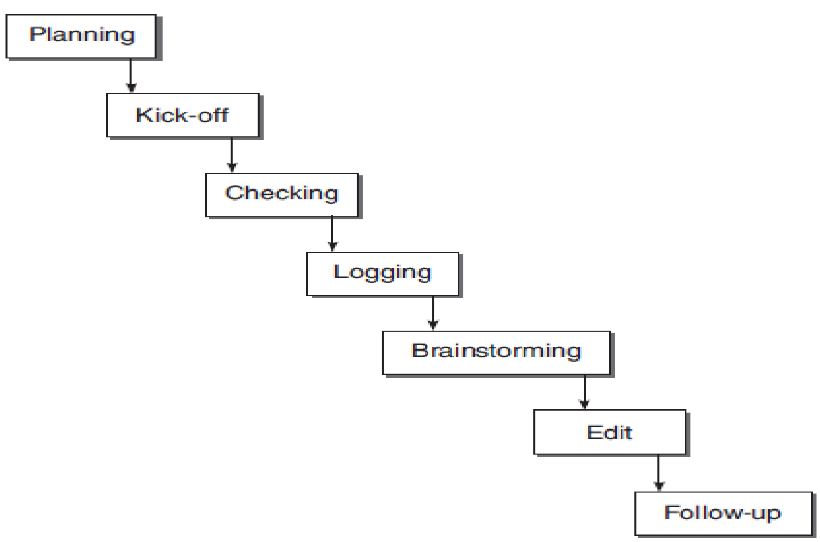
- If required, any unresolved issues are discussed in this step.

Conclusion

- The final report of the inspection process along with the analysis is produced by the moderator.

Gilb Inspection

- Gilb and Graham defined this process Three different roles are defined in this type of inspection:
  - **Leader** is responsible for planning and running the inspection.
  - **Author of the document**
  - **Checker** is responsible for finding and reporting the defects in the document
- The inspection process consists of the following steps, as shown below:



Gilb inspection process

Entry

- The document must pass through an entry criteria so that the inspection time is not wasted on a document which is fundamentally fl awed.

Planning

- The leader determines the inspection participants and schedules the meeting.

Kick-off

- The relevant documents are distributed, participants are assigned roles and briefed about the agenda of the meeting.

Checking

- Each checker works individually and finds defects.

Logging

- Potential defects are collected and logged.

Brainstorm

- In this stage, process improvement suggestions are recorded based on the reported bugs.

Edit

- After all the defects have been reported, the author takes the list and works accordingly.

Follow-up

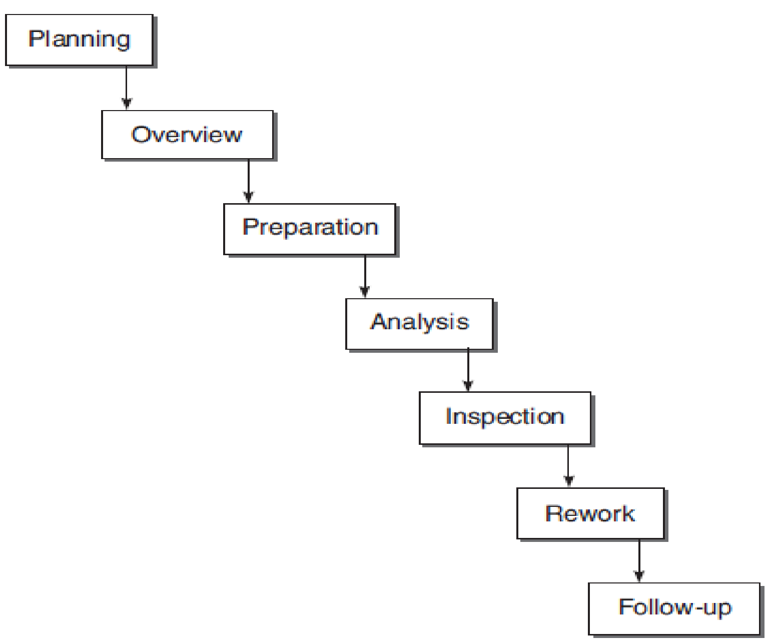
- The leader ensures that the edit phase has been executed properly.

Exit

- The inspection must pass the exit criteria as fi xed for the completion of the inspection process.

Humphrey's Inspection Process

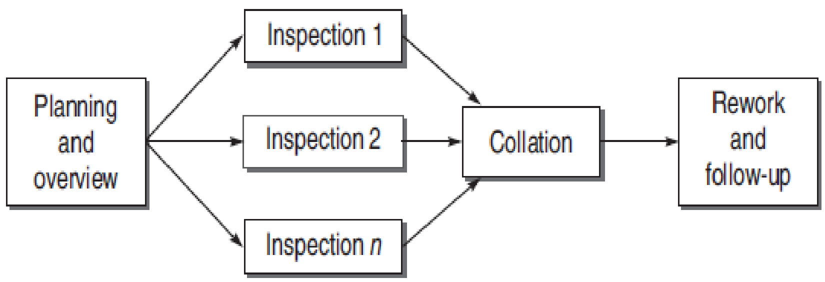
- In this process, the preparation phase importanc on finding and logging of bugs.
- It also includes an analysis phase between preparation and meeting.
- In the analysis phase, individual logs are analysed and combined into a single list.



Humphrey's process

N-Fold Inspection

- It is based on the idea that the effectiveness of the inspection process can be increased by increasing the number of teams inspecting the item, the percentage of defects found may increase.
- So this process consists of many independent inspection teams.
- This process needs a coordinator who coordinates various teams, collects and collates the inspection data received by them. For this purpose, a moderator is assigned of every inspection team.
- This process consists of the following stages:



N-fold inspection

Planning and overview

- This is the formal planning and overview stage, it includes the planning of how many teams will participate in the inspection process.

Inspection stages

- There are many inspection processes, It is not necessary that every team will choose the same inspection process.

Collation phase

- The results from each inspection process are gathered, collated, and a master list of all detected defects is prepared.

Rework and follow-up

- This step is same as the tradition Fagan inspection process.

Phased Inspection

- Phased inspections are designed to verify the product in a particular domain.
- If we feel that in a particular area, we may encounter errors, then phased inspections are carried out, only experts who have experience in that particular domain are called.
- In this process, inspection is divided into more than one phase. There are two types of phases

Single inspector

- In this phase, a rigorous (strict) checklist is used by a single inspector to verify whether the features specified are there in the item to be inspected.

Multiple inspectors

- Here, a checklist cannot be used. The item cannot be verified just with the questions mentioned in the checklist.
- There are many inspectors, each inspector examines this information and develops a list of questions of their own based on a particular feature.
- The item is then inspected individually by all the inspectors based on a self-developed checklist, after a reconciliation meeting is organized where inspectors compare their findings about the item.

1.1.7 Reading Techniques

- A reading technique can be defined as a series of steps or procedures whose purpose is to guide an inspector to acquire a deep understanding of the inspected software product.
- A reading technique is a mechanism for the individual inspector to detect defects in the inspected product. The various reading techniques are:

Ad hoc method

- The word *ad hoc* only refers to the fact that no technical support on how to detect defects in a software
- In this case, defect detection fully depends on the skills, knowledge, and experience of an inspector.

Checklists

- A checklist is a list of items that focus the inspector's attention on specific topics, such as common defects or organizational rules, while reviewing a software document.

Scenario – Based Reading

- Different methods developed based on scenario based reading are:

Perspective-based reading

- A software item should be inspected from the perspective of different stakeholders.
- Inspectors of an inspection team have to check the software quality as well as the software quality factors of a software from different perspectives.
- The perspectives mainly depend upon the roles people have within the software development.
- For example, a testing expert first creates a test plan based on the requirements specification and then attempts to find defects from it.

Usage-based reading

- This method is applied in design inspections.
- Design documentation is inspected based on use cases, which are documented in requirements specification.

Abstraction driven reading

- This method is designed for code inspections.
- In this method, an inspector reads a sequence of statements in the code and extracts the functions these statements compute.

Task-driven reading

- This method is also for code inspections.
- In this method, the inspector has to create a data dictionary, a complete description of the logic and a cross-reference between the code and the specifications.

Function-point based scenarios

- This is based on scenarios for defect detection in requirements documents by function-point analysis (FPA).
- A Function Point Scenario consists of questions and directs the focus of an inspector to a specific function-point item within the inspected requirements document.

1.1.8 Checklists for Inspection Process

- The inspection team must have a checklist against which they detect errors. The checklist is according to the item to be inspected.
- Checklists should be prepared in consultation with experienced staff and regularly updated as more experience is gained by the inspection process.
- The checklist may vary according to the environment and needs of the organization. Each organization should prepare its own checklists for every item.

1.2 Structured Walkthroughs

- It is a less formal and less rigorous technique as compared to inspection.
- The very common term used for static testing is Inspection but it is for very formal process.
- If you want to go for a less formal having no bars of organized meeting, then walkthroughs are a good option.
- The inspection team uses the checklist approach for uncovering errors. A walkthrough is less formal, has fewer steps and does not use a checklist to guide.
- Rather than simply reading the program or using error checklists.
- Structured walkthrough team consists of the following members:

Coordinator Organizes, moderates, and follows up the walkthrough activities.

Presenter/Developer Introduces the item to be inspected.

Scribe/Recorder Notes down the defects found and suggestion proposed by the members.

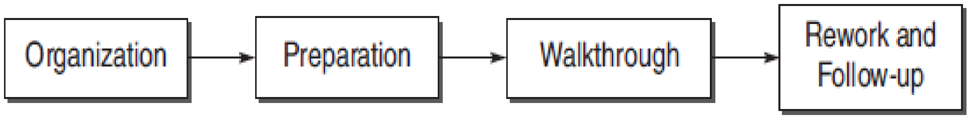
Reviewer/Tester Finds the defects in the item.

Maintenance Oracle Focuses on long-term implications and future maintenance of the project.

Standards Bearer Assesses commitment to standards.

User Representative/Accreditation Agent Reflects the needs and concerns of the user.

- The steps of a walkthrough process are shown below:



Walkthrough process

1.3 Technical Reviews

- A technical review evaluate the software in the light of development standards, guidelines, and specifications and to provide the management with evidence that the development process is being carried out according to the stated objectives.
- A review is similar to an inspection or walkthrough, except that the review team also includes management.
- Therefore, it is considered a higher-level technique as compared to inspection or walkthrough.
- A technical review team is generally comprised of management-level representatives and project management.
- Review agendas should focus less on technical issues and more on oversight than an inspection.